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Energy Market



Global Energy Resources - Data

- ♦ Data is the basis of various decisions
 - environmental
 - political
 - economic
- ♦ Official data sources
 - Statistic Austria (STAT)
 - Statistical Office of the European Commission (Eurostat)
 - International Energy Agency (IEA)

The official data sources are harmonized

Global Energy Resources – Economical Growth

- ♦ Energy always played an important role
- ♦ Economical growth is linked to
 - Cheap energy
 - Reliable energy
 - Easy-to-use

Global Energy Resources – Energy GDP Nexus

- ♦ Link between Energy Consumption <-> Energy growth
 - Energy Gross Domestic Product (GDP) Nexus

- ♦ Four hypothesis
 - Growth hypothesis – Energy consumption drives economic growth
 - Conservation hypothesis – High economic growth drives energy consumption
 - Feedback hypothesis – Bidirectional relationship: energy and gross domestic product (GDP dt: Bruttoinlandsprodukt) influence each other
 - Neutrality hypothesis – no relationship: energy consumption and economic growth are independent of each other

Global Energy Resources – Energy GDP Nexus

- ♦ General rule: Cheap and easy accessible energy -> adv. for manufacturing companies
- ♦ Exeptions: Nigeria, Venezuela – favorable energy condition – no GDP increase

Global Energy Resources - Classification

- ♦ Stages of energy flow
 - Primary Energy
 - Transformation Energy
 - Secondary Energy
 - Final Energy Consumption (dt. Endenergie)
 - Useful Energy (dt. Nutzenergie)

Global Energy Resources - Classification

- ◆ Primary Energy
 - Energy commodities (dt. Rohstoffe) extracted or captured directly from natural resources
- ◆ Examples
 - Water
 - Natural gas
 - Crude oil
 - Coal
 - Uranium (Nuclear thermal power)
 - Sunlight
 - Wind
 - ...

Global Energy Resources - Classification

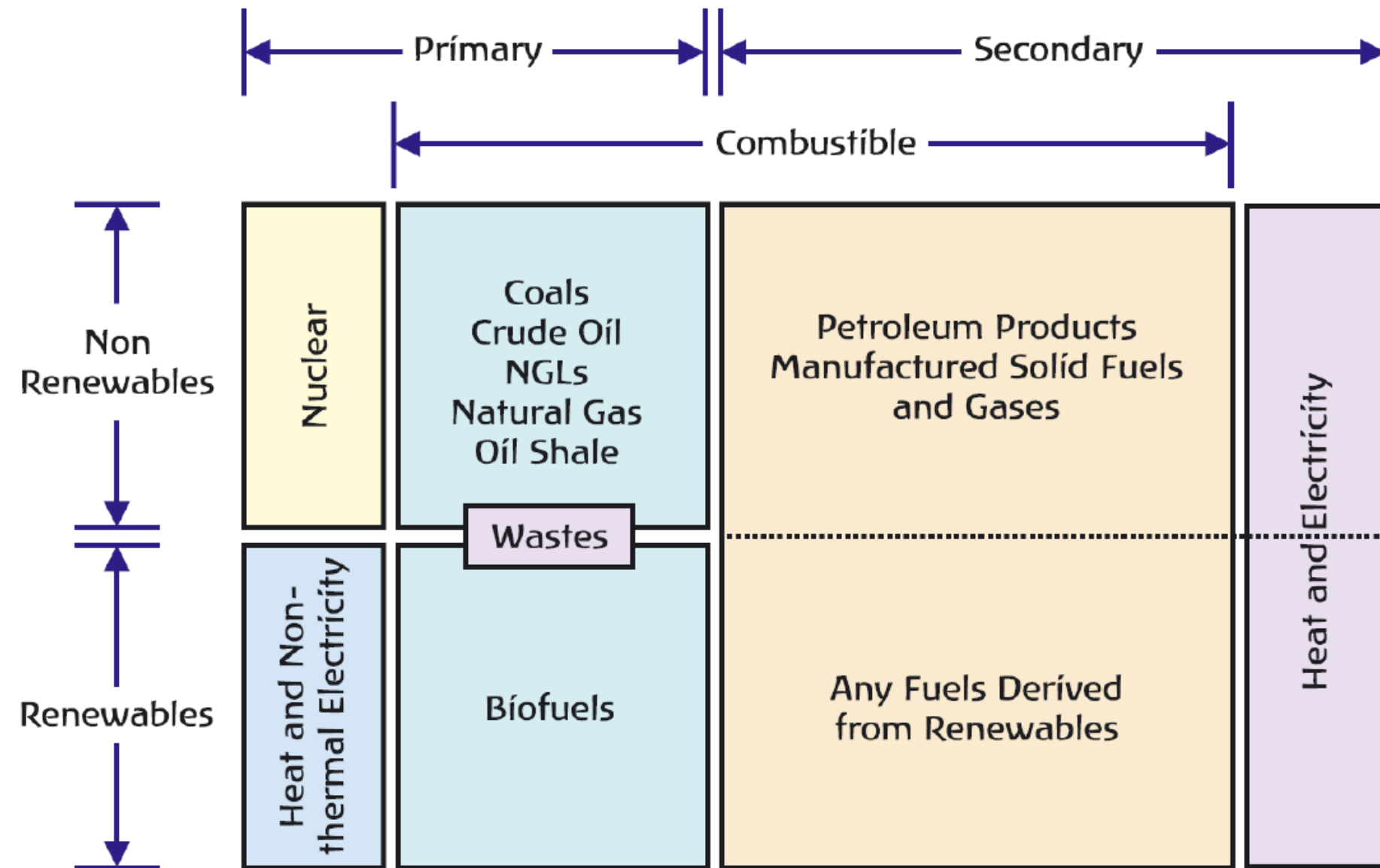
- ♦ Secondary Energy
 - Primary Energy transformed into another energy form
- ♦ Example
 - Electricity
- ♦ Final Energy Consumption
 - Energy made available for consumption
- ♦ Useful Energy
 - Air conditioning
 - IT

Global Energy Resources - Classification

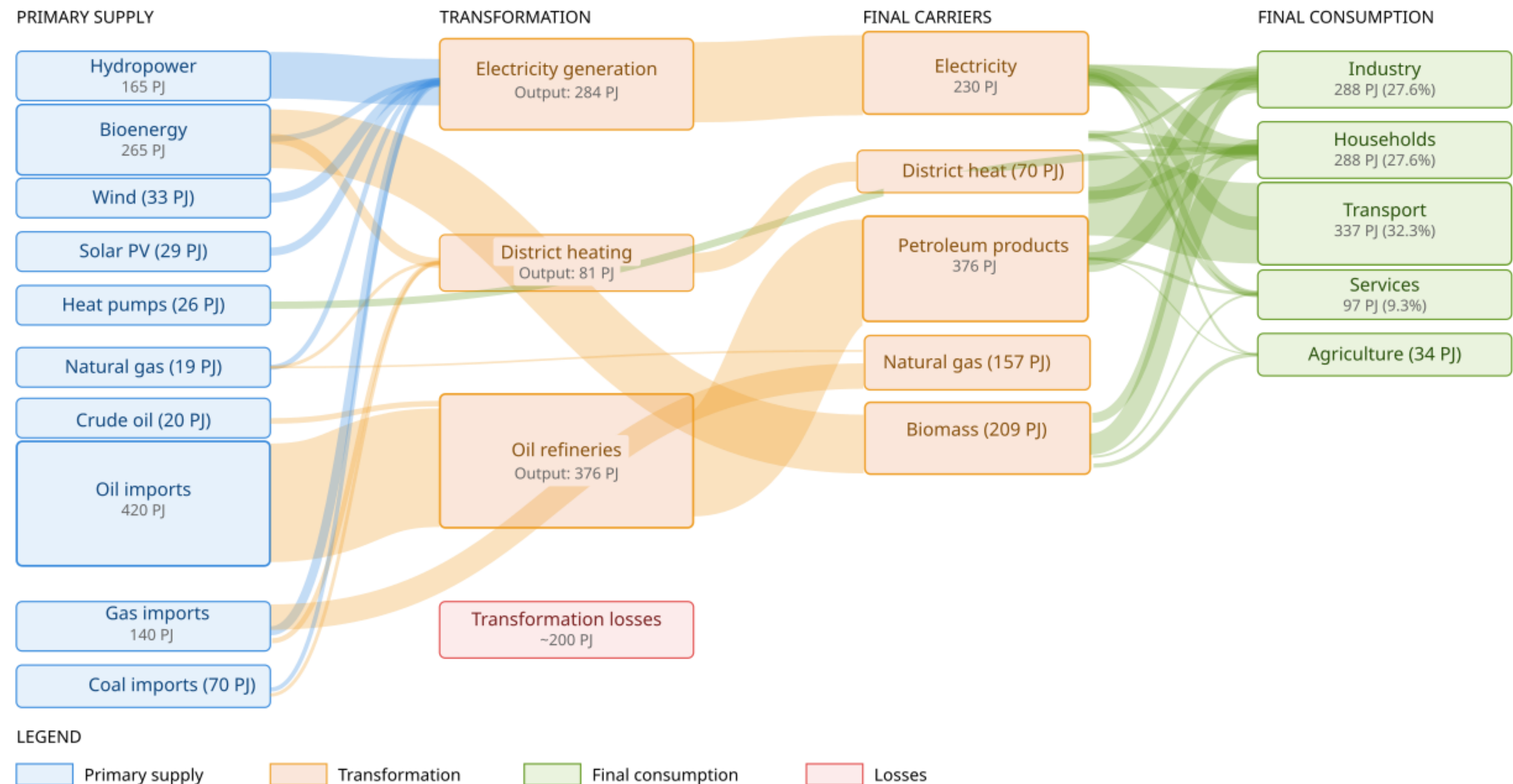
- ♦ Example: Electricity is generated by burning oil
 - Oil is primary energy
 - Electricity is secondary energy
- ♦ That electricity is transmitted to a factory where the energy is used for cooling
 - The energy output from the air conditioner is the useful energy

Global Energy Resources - Classification

- ♦ Primary / Secondary
- ♦ Renewables / Non Renewables
- ♦ Non-Renewables
 - Fossil fuels
 - Nuclear



Global Energy Resources – Energy Flow Austria



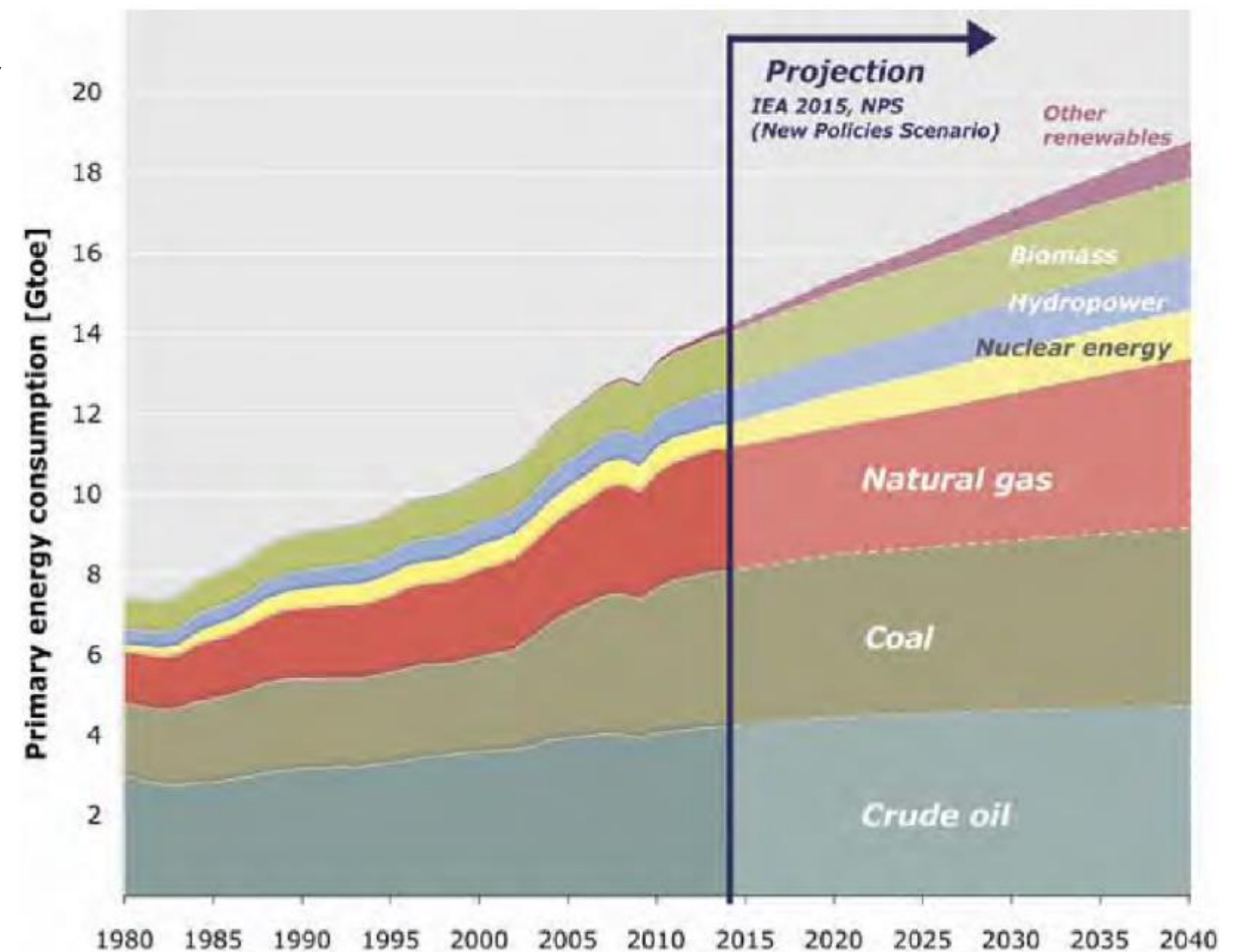
Definitions

- ♦ Reserve: Proven amounts which can economically produced with today's technology
- ♦ Resources: estimated or proven potential amounts which can currently not economically produced with today's technology

- ♦ Exa = 10^{18}
- ♦ Exa = 10^{15}
- ♦ Mega = 10^6
- ♦ Giga = 10^9

Primary Energy

- ♦ Growing renewable
 - hydro-electric contributes significantly
 - All others only 2.5 % share
- ♦ Dominance of non-renewable
- ♦ Energy Reserve: 37 934 Exa Joule
- ♦ Energy Resources: 551 813 EJ

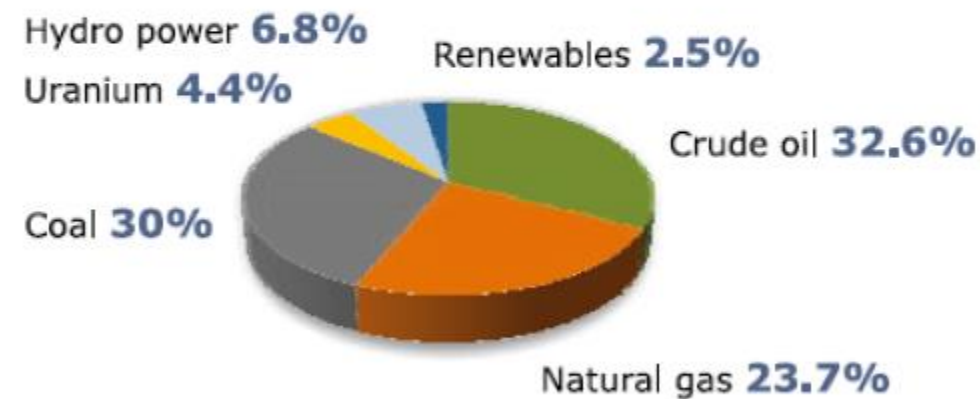


toe: tonne of oil equivalent
1 toe = 42 GJ

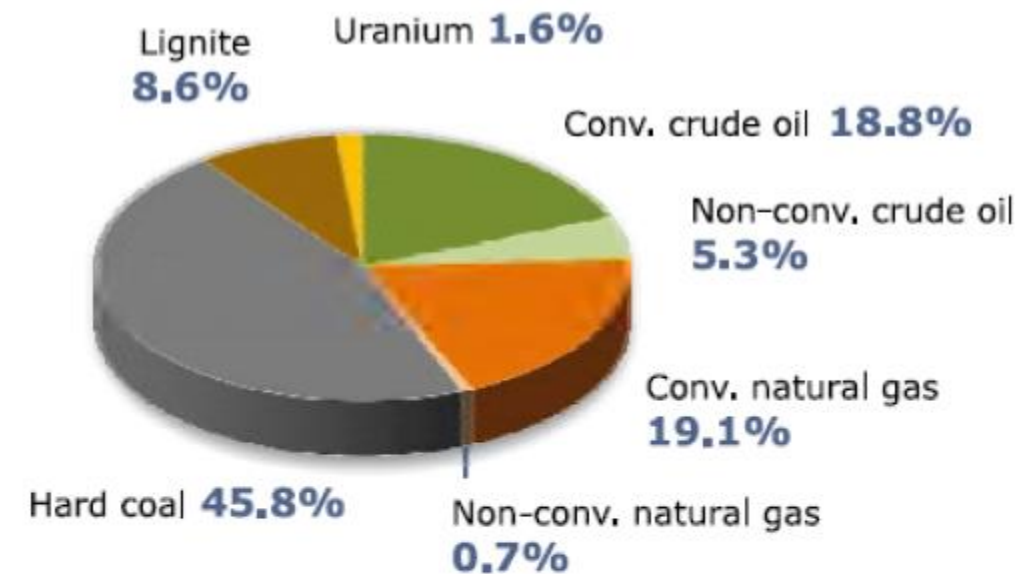
Primary Energy

- ♦ How long can we continue with the current energy consumption / production?

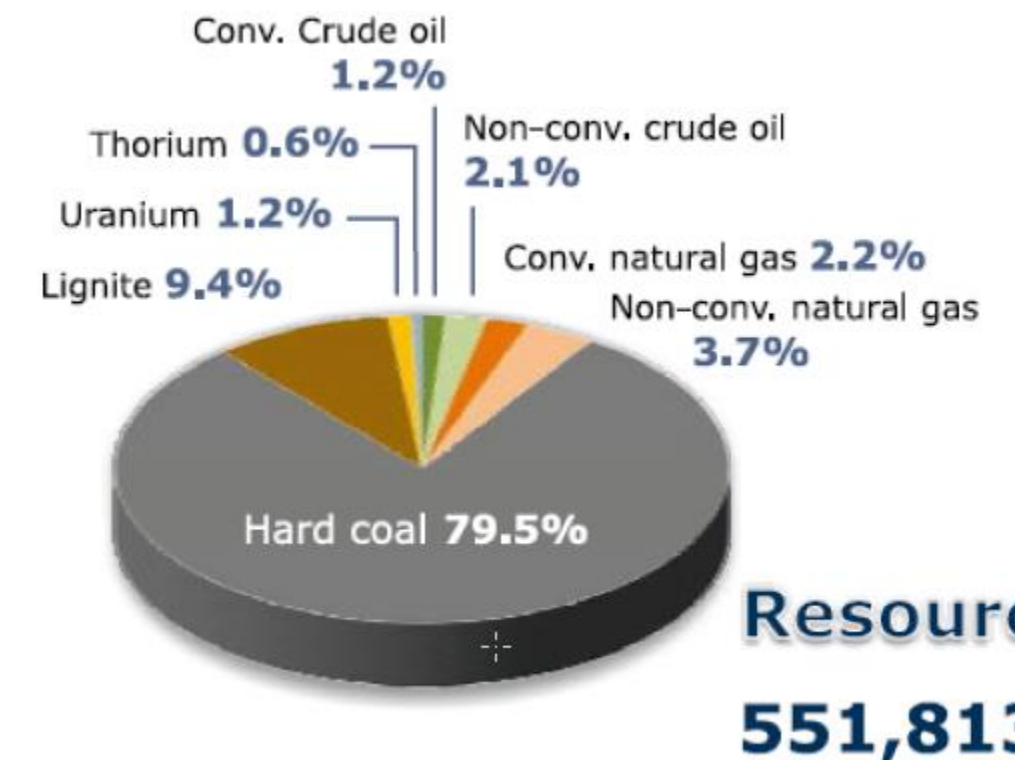
Energy consumption 540 EJ



Production 522 EJ



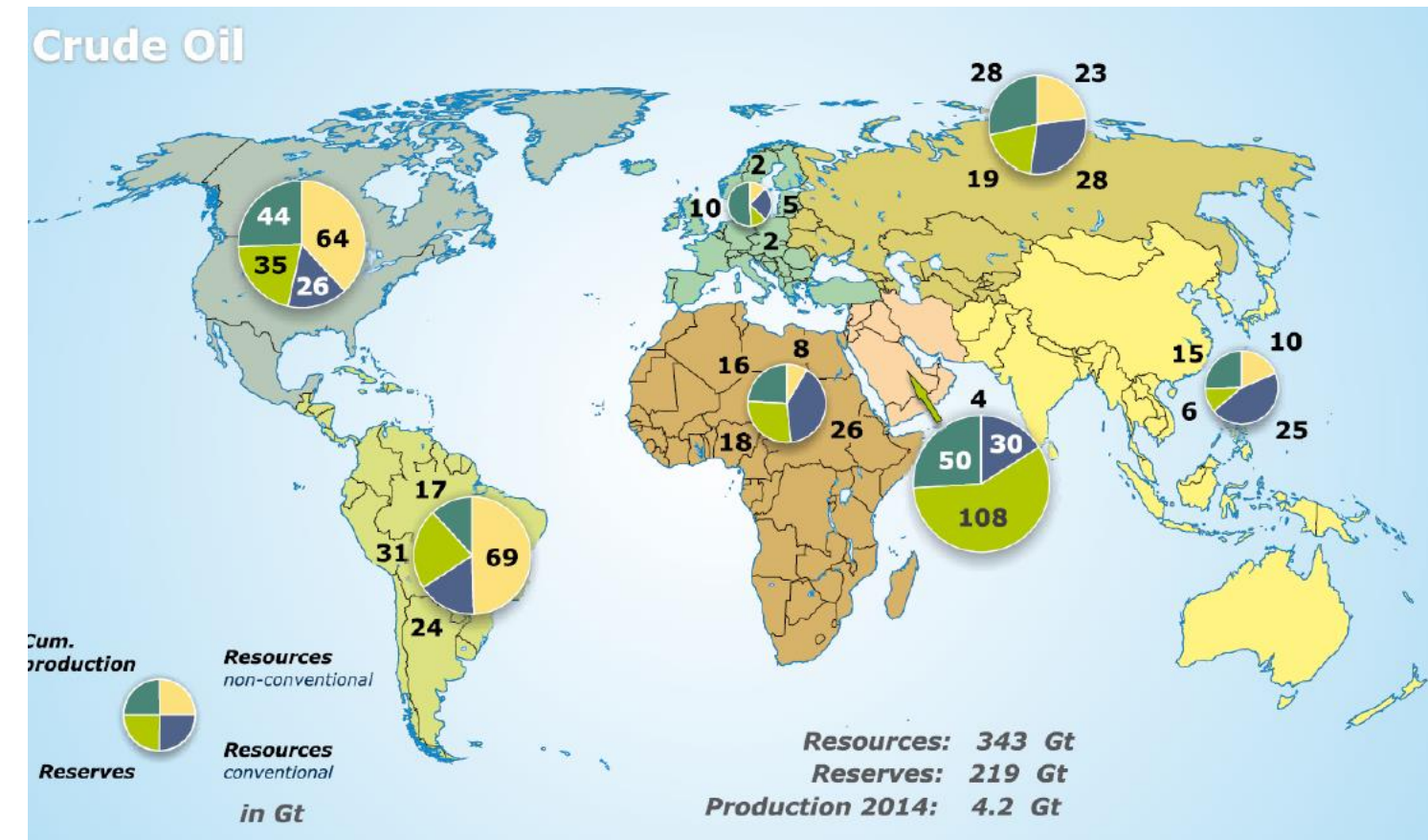
Reserves 37,934 EJ



Resources 551,813 EJ

Primary Energy – Crude Oil

- ◆ Most important primary source (32%)
- ◆ 60% held by Saudi Arabia, Canada, Venezuela, Iran and Irak
- ◆ Shale oil (crude oil from rocks) caused a production boom in the US
- ◆ Production in Europe declined (0.6%)
- ◆ 3-5% increase in CIS, Africa, Middle East, NA, LA



Primary Energy – Crude Oil

- ♦ 20 largest consumer countries use 75 % of available mineral oil
- ♦ Austria has also a crude oil production

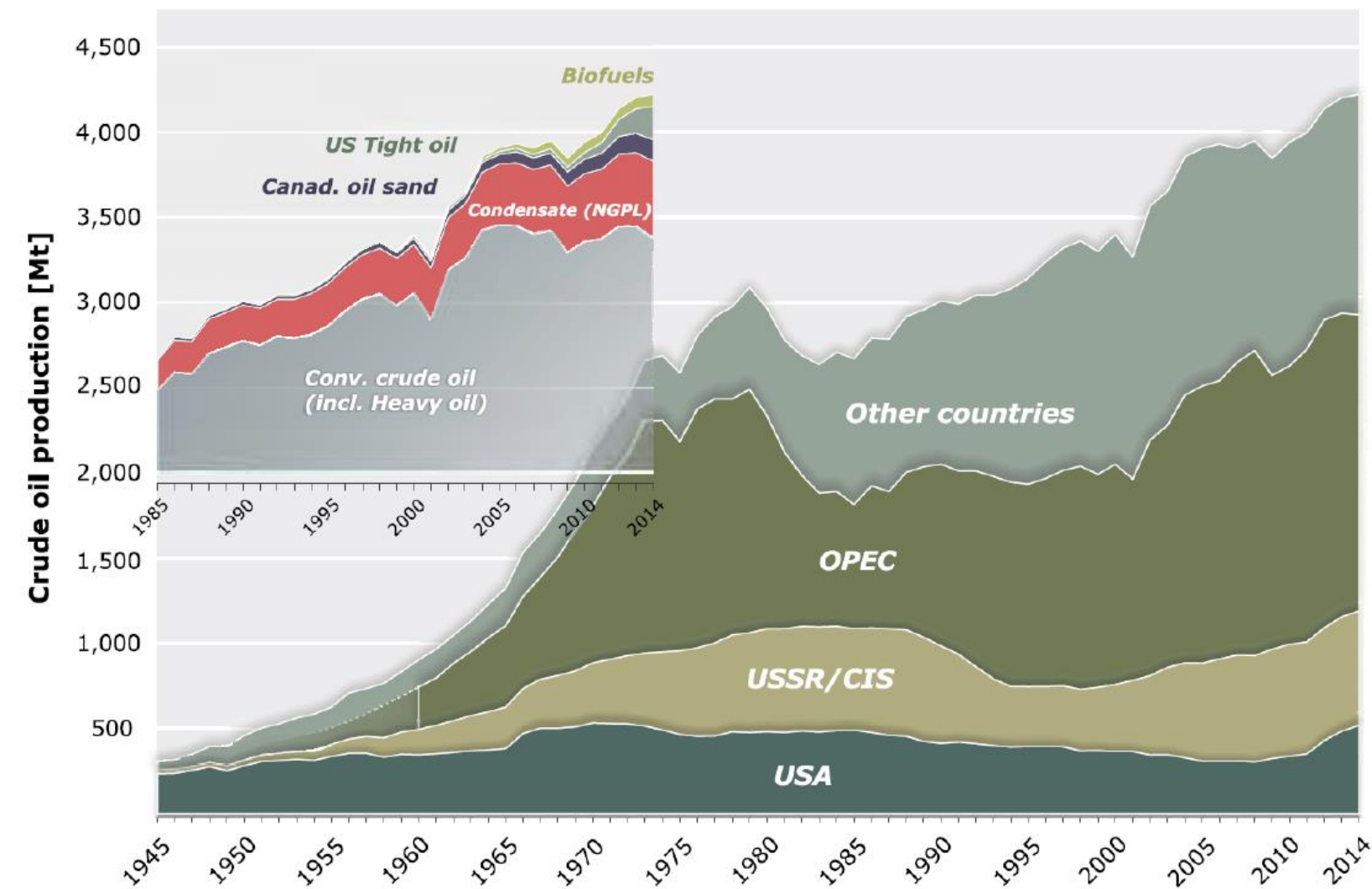
Country	Production	Cum. Production	Reserves	Resources
Austria	0.5	128	5	10
Germany	1.9	315	18	240
Italy	5.4	219	80	1540
Norway	93.7	4202	1057	2415
UK	48.5	3913	340	1263

Table 2.1.: Excerpt of the crude oil production 2020 [Mt] - Europe.

Source: [4]

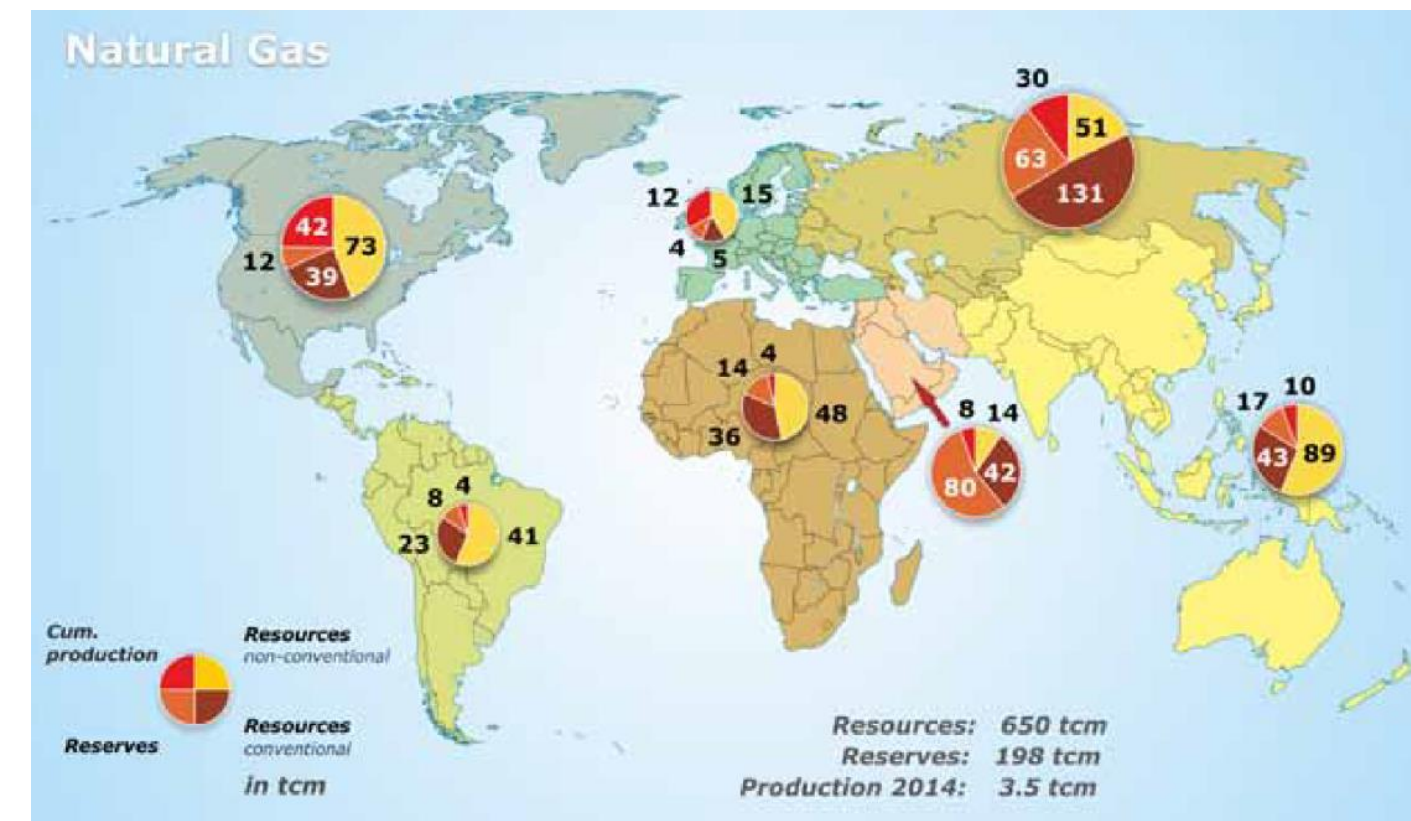
Primary Energy – Crude Oil

- ◆ 1945 – US largest producer
- ◆ 1960 – OPEC became biggest producer (Headquarters is in Vienna)
- ◆ OPEC: Saudi Arabia, Venezuela, Iran, Iraq, Kuwait, United Arab Emirates, Qatar, Nigeria, Libya, Indonesia, Ecuador, Angola, Algeria
- ◆ 1973/74 and 1979/89 – Oil crisis
- ◆ Soviet Union – 1960 onward expansion until 1990. 2000 onward (Vladimir Putin) start expansion again
- ◆ 2008 – US expands production large-scale crude oil from tight reservoir rocks



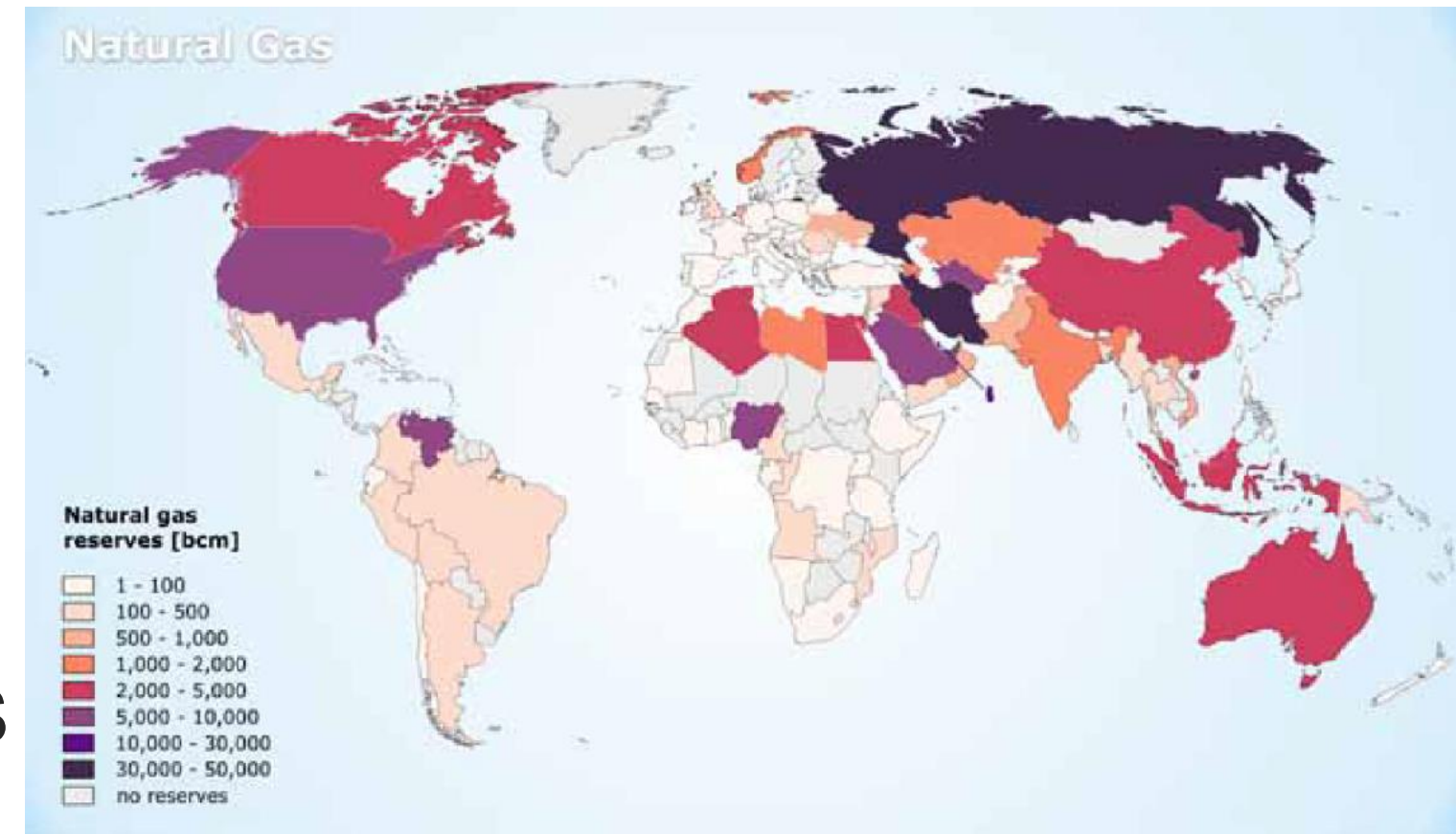
Primary Energy – Natural Gas

- ◆ Third most important primary energy resource (23,7 %)
- ◆ US covers 95% of its domestic demand
- ◆ Trend to liquefied natural gas (LNG) – enables storage and transport (trading)
- ◆ Global consumptions increased (1.4%)
- ◆ EU consumption decreased by 10% (mild winter)
- ◆ Geopolitical risk factor in the reliable supply of natural gas



Primary Energy– Natural Gas

- ♦ Russia largest conventional natural gas resources
- ♦ Shale gas reserve in US (46% of their reserve)
- ♦ Russia - More than half of natural gas reserve (54%)
- ♦ 80% of global reserves are in OPEC and CIS countries



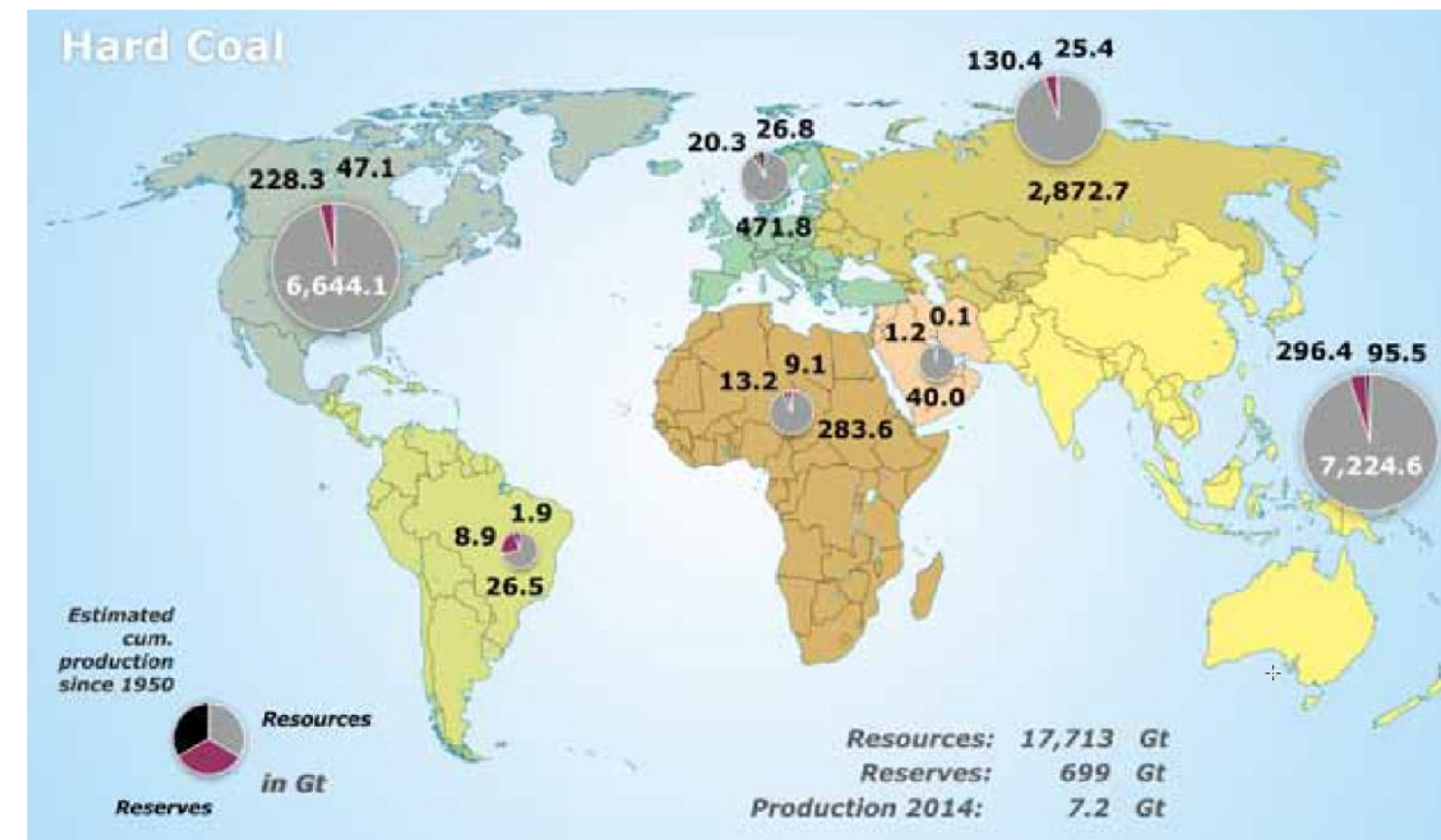
Primary Energy– Natural Gas – Production

Rank	Country	2020
1	USA	948
2	Russia	693
3	Iran	254
4	China	205
5	Qatar	185
6	Canada	158
7	Australia	154
8	Norway	110
9	Saudi Arabia	103
10	Algeria	85

Table 2.2.: Top 10 natural gas producer 2020 [Mrd. kubic meter]
Source: [4]

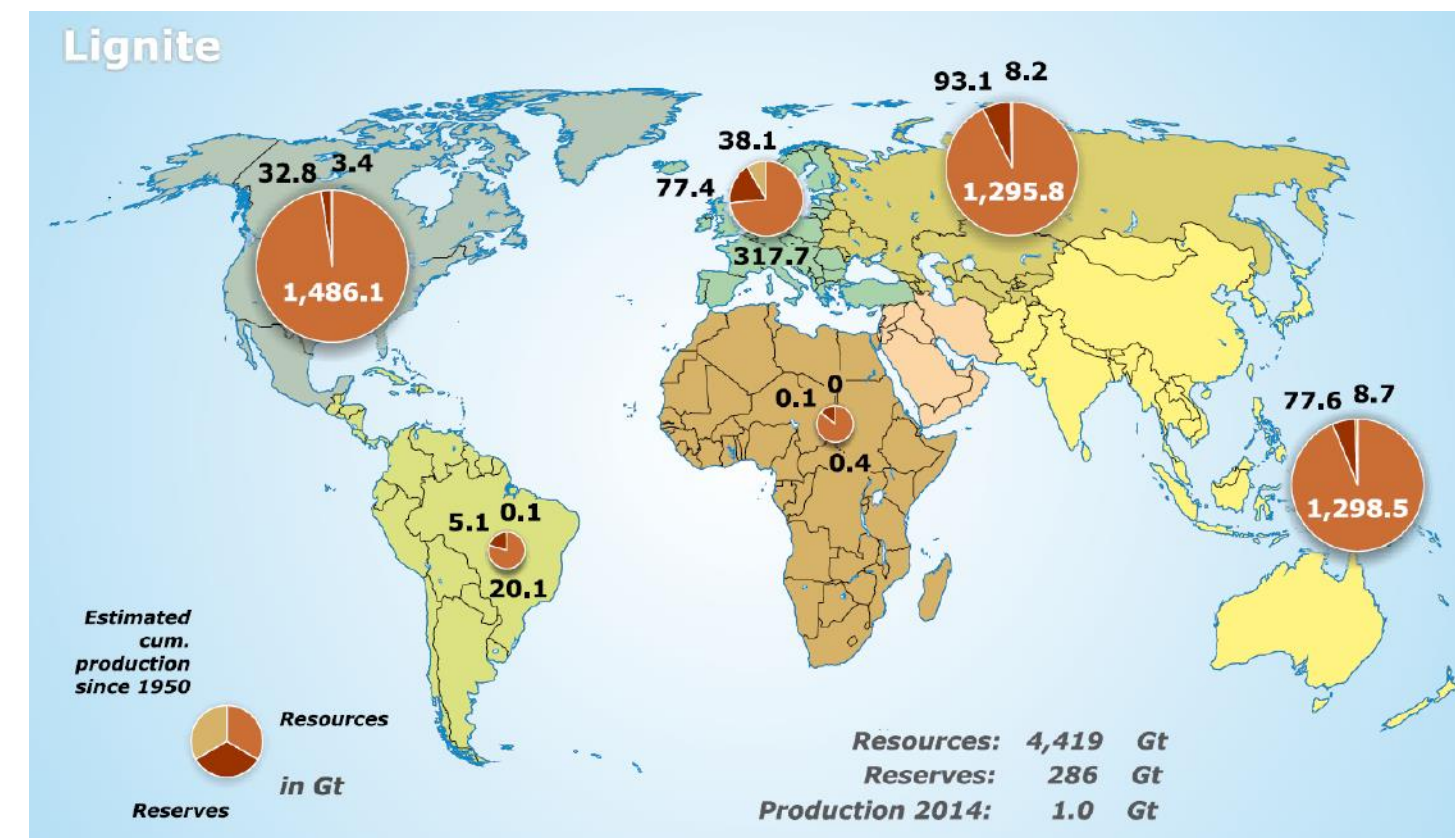
Primary Energy – Coal – Hard Coal (dt. Steinkohle)

- ♦ 30 % global share
 - 2nd largest primary energy resource
- ♦ Coal has the most resources from all fossil energies – will last the longest
- ♦ Hard coal: energy content 16 500 kJ/kg
 - cheaper to transport and is traded
- ♦ Largest producer China (52.1 %), followed by US (11,7%) and India (8,6%)
- ♦ EU share 1,5 %
- ♦ China needs half of the global hard coal demand



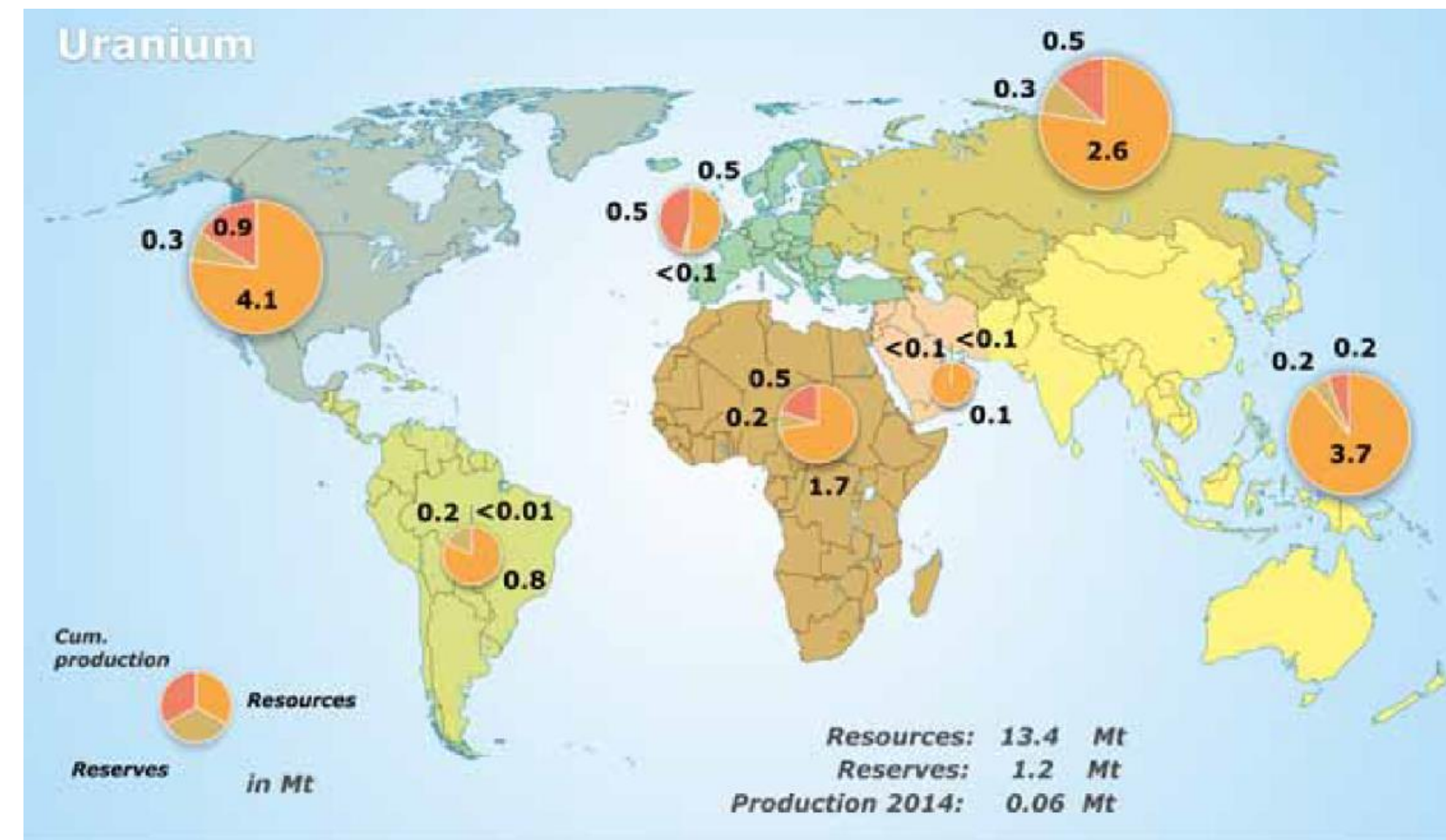
Primary Energy – Coal – Lignite (dt. Braunkohle)

- ♦ Lignite is too expensive to transport
- ♦ Mainly used for electricity production
- ♦ NA has largest resources of lignite
- ♦ Russia has biggest share of reserves (Germany 12,7%)



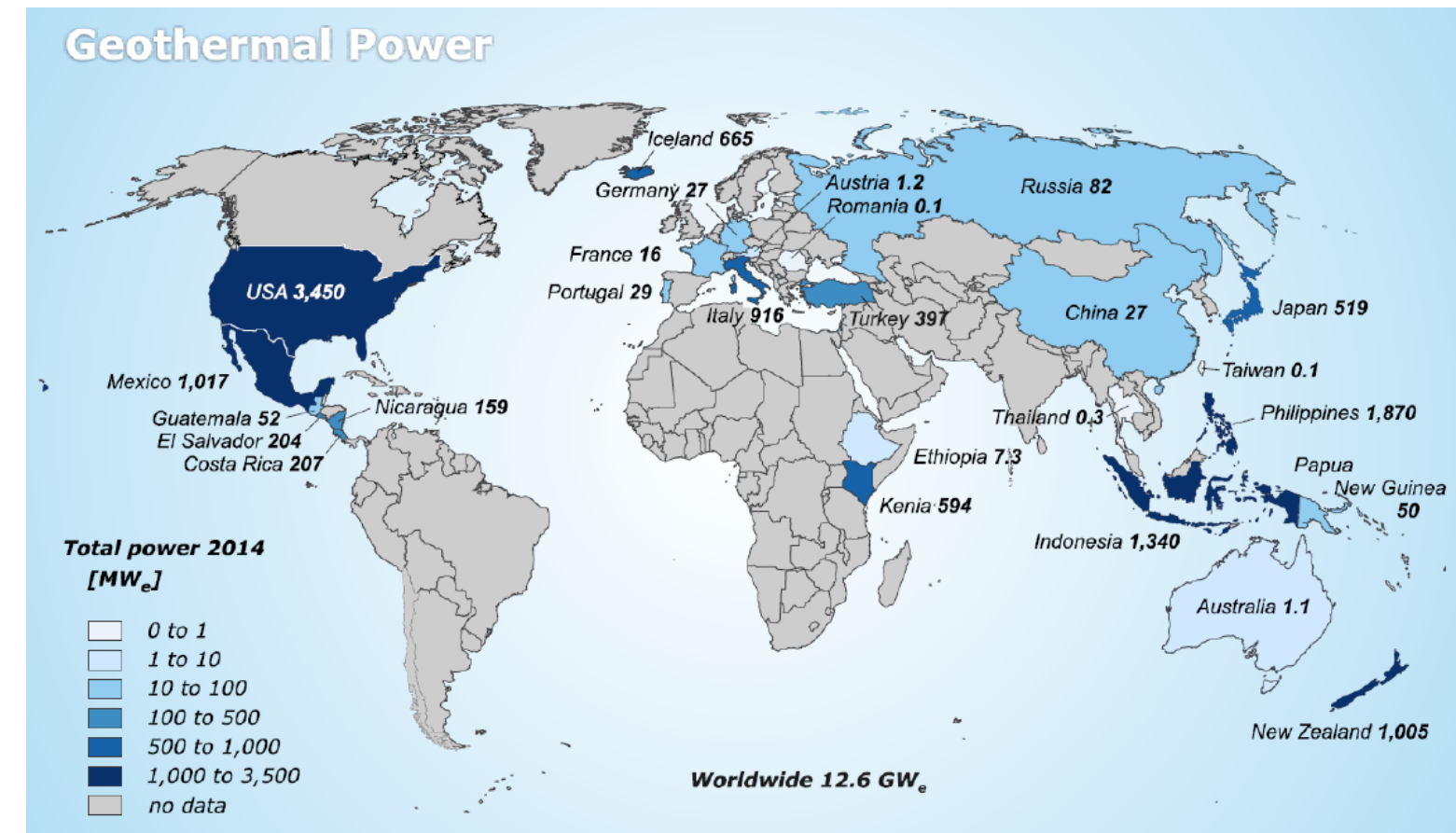
Primary Energy - Uranium

- ◆ 83% of global production by eight mining companies
- ◆ Half of the demand for US, France and China
- ◆ Growth in demand expected
- ◆ At end of 2014 70 nuclear power plants were in construction
 - China 26, Russia 9, India 6, ...
- ◆ In Germany eight nuclear power plants have been shut down in 2011



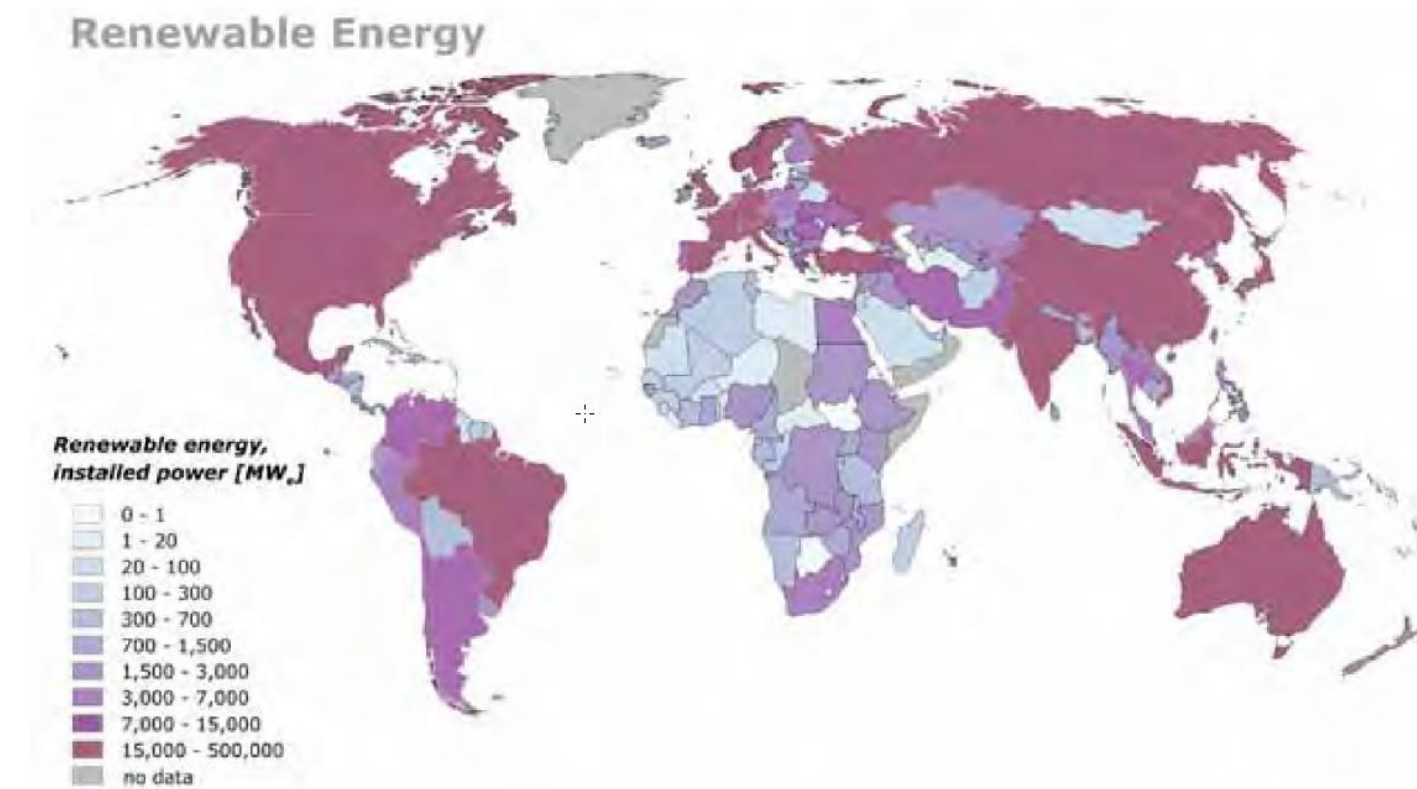
Primary Energy – Deep Geothermal Energy

- ◆ Definition
 - wells deeper than 400 m
 - Temperatures exceeding 20°C
- ◆ Development depends on
 - national objectives
 - political and social frameworks



Primary Energy – Renewable Energy

- ♦ High variety
 - Solid biomass (e.g. wood)
 - Hydroelectric power
 - Wind power
 - Solar power
 - Geothermal energy
- ♦ 65% Hydro electric
- ♦ 20% Wind Power
- ♦ 10% Photovoltaics
- ♦ China is production leader, followed by US, Brazil and Germany





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